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An Interpretable Ensemble Learning Framework for Leakage Localization in Urban Water Distribution Networks

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Abstract

Rapid and accurate leakage localization in urban water distribution networks (WDNs) is vital for reducing non-revenue water and supporting sustainable water resource management. This study develops a spatial–hydraulic coupled and interpretable machine learning framework for zonal leakage localization under operational uncertainties. The WDN is first partitioned into zones, followed by leakage-region prediction using a Stacking ensemble integrating Random Forest, XGBoost, and MLP. SHAP analysis is employed to characterize the contribution of individual monitoring nodes to regional classification. The proposed Stacking framework achieves an overall localization accuracy of 0.9667 and a Macro-F1 score of 0.9666, with relatively stable diagnostic performance under pressure noise. Global SHAP analysis identifies several monitoring nodes with substantial contributions to the overall classification, while the observed misclassifications are mainly concentrated between a small number of neighboring regions with similar feature representations. A further test on the Net3 network achieves an overall localization accuracy of 0.9812, providing additional evidence of the applicability of the framework across different network configurations. Overall, the results indicate that the proposed framework provides an accurate and interpretable approach for leakage localization in WDNs.

Keywords: WDNs; SHAP; leakage localization; ensemble learning; model interpretability

1. Introduction

A water distribution network (WDN) is an engineering system composed of water sources, pipelines, pumping stations, and storage facilities. Its primary function is to deliver treated water resources to various users according to demand. Such systems play a critical role in ensuring the continuity of water supply and water quality safety, and are of great importance to public health, urban operation, and emergency response. During long-term operation, WDNs are subjected to various adverse factors, such as pipeline aging, corrosion, external load variations, and construction disturbances. These factors gradually weaken the structural strength and sealing performance of pipelines, thereby increasing the probability of leakage occurrence within the network. According to the development characteristics of leakage, water network leakage can generally be classified into two categories: burst leakage and progressive (hidden) leakage [1]. Burst leakage refers to leakage events caused by sudden structural failure of pipelines within a very short period of time, such as pipe bursts, joint fractures, or external damage. This type of leakage is typically characterized by its abrupt occurrence, large leakage flow rate, and significant disturbance to the hydraulic state of the system, often accompanied by rapid pressure

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drops and abnormal flow fluctuations within the network. In contrast, progressive (hidden) leakage develops gradually as a result of pipeline aging, corrosion, or deterioration of sealing performance. At the early stage, the leakage rate is relatively low and has limited influence on hydraulic parameters, making it difficult to trigger obvious pressure or flow anomalies; however, the leakage severity tends to increase progressively over time. Once leakage occurs in a WDN, it not only leads to water resource waste and reduced water supply efficiency, but may also cause abnormal network pressure and secondary water contamination, thereby affecting water supply safety and the operational stability of the system [2]. Therefore, in order to reduce water losses and operational risks caused by leakage and to enhance the safety and stability of WDNs, it is of great significance to conduct effective research on leakage detection and localization.

Previous studies on leakage detection have mainly focused on three categories of methods: model-based methods, statistical methods, and machine learning methods [3–5]. Model-based methods establish hydraulic models of WDNs to simulate the expected sensor readings under normal operating conditions, and then compare these predicted values with the actual monitoring data collected by Supervisory Control and Data Acquisition (SCADA) systems. When significant discrepancies (residuals) occur between the predicted and measured values, these residuals can be used to determine the existence of leakage within the system and further infer the leakage location based on their spatial distribution characteristics. For example, Keramat, et al. [6] proposed a matched-filter objective function combined with the inverse transient method to detect leakage in WDNs under measurement noise conditions. Levinas, et al. [7] developed a novel method based on transient modeling of water networks, capable of generating high-resolution pressure responses under various scenarios to identify leaking pipes in pressurized WDNs. Vrachimis, et al. [8] proposed a leakage detection and localization approach using interval models, which achieved higher detection rates compared with similar methods employing bounded uncertainties. Although these model-based approaches demonstrate strong theoretical rigor, hydraulic models are often difficult to accurately establish and continuously update in practical applications. In addition, uncertainties in parameters such as pipe roughness, pipe diameter, and nodal water demand, together with the high computational cost, limit their applicability in large-scale and real-time scenarios. Furthermore, some high-performing model-driven approaches rely heavily on an in-depth understanding of the hydraulic characteristics of WDNs and high-quality model calibration, conditions that are often difficult to satisfy in real-world applications. Hydraulic transient analysis also requires extremely high temporal resolution of monitoring data, typically at the second or even millisecond level, which existing SCADA systems generally cannot provide. This mismatch further constrains the practical implementation of such methods [4,9–11].

Statistical methods reduce the dependence on hydraulic models and alleviate the burden associated with model calibration by directly analyzing variations and anomalies in observed operational data, such as flow and pressure measurements collected from SCADA systems. Among these approaches, statistical process control (SPC) methods are widely applied for leakage detection due to their simplicity, computational efficiency, and suitability for real-time monitoring. Representative SPC techniques include the Western Electric Company rules [12], cumulative sum (CUSUM) method [13], exponentially weighted moving average (EWMA) method [14,15], and Hotelling's T^2 method [16]. Specifically, rule-based approaches identify abnormal operating conditions by detecting deviations from predefined thresholds, while CUSUM and EWMA methods improve sensitivity to small or persistent changes by accumulating and smoothing measurement deviations. Hotelling's T^2 method extends statistical monitoring to multivariate scenarios by considering correlations among multiple measurements. In addition, Bayesian theory-based statistical approaches

have also been investigated for leakage detection, where prior knowledge and probabilistic inference are incorporated to quantify uncertainties and improve decision-making under incomplete information [17]. Despite these advantages, statistical methods still face several limitations. Most conventional statistical approaches rely on assumptions that SCADA data follow known or approximately Gaussian distributions, which may not hold under complex operating conditions [4]. Moreover, their detection performance can be affected by varying demand patterns, sensor noise, and other non-leakage-related disturbances. These limitations restrict their ability to capture highly nonlinear relationships and complex interactions among multiple influencing factors.

Machine learning methods have demonstrated strong capabilities in automatically learning complex patterns from large-scale and high-dimensional datasets, capturing nonlinear relationships among multiple variables, and adapting to diverse data characteristics. Consequently, they have been increasingly applied across various domains and have gradually become prominent approaches in the field of leakage detection in recent years. For example, Zhang, et al. [18] employed a multi-class support vector machine to identify leakage zones in WDNs. El-Zahab, et al. [19] developed a leakage detection model using accelerometers and analyzed it with support vector machines, decision trees, and naïve Bayes, achieving promising results. Bermúdez, et al. [20] utilized the k-nearest neighbors classification algorithm to identify leakage-affected regions in WDNs. Guo, et al. [21] represented leakage signal characteristics using leakage spectrograms and developed a time-frequency convolutional neural network model for leakage signal identification. Pérez-Pérez, et al. [22] designed a two-stage neural network in which the first stage calculated friction factors from inlet and outlet flow rates, while the second stage utilized this information together with inlet and outlet pressures to localize leakage positions. Cody Roya, et al. [23] combined convolutional neural networks with variational autoencoders to propose a semi-supervised method for detecting leakage in WDNs. Liao, et al. [24] applied deep learning techniques to identify leakage in WDNs through transient frequency responses.

Generally, leakage localization relies on monitoring abnormal hydraulic or physical responses caused by pipe failures. Traditional approaches often employ specialized sensing devices, such as acoustic instruments, to identify leakage locations by capturing abnormal sound signals generated around suspected leakage points [25]. Although acoustic-based techniques can provide direct leakage information, their practical application is limited by the requirement for extensive field inspections and the considerable time consumption associated with surveying numerous potential leakage sections. To overcome these limitations, data-driven approaches based on measurements collected from sensor networks have been increasingly investigated for leakage detection and localization. Among various monitoring signals, pressure and flow measurements are the most commonly used data sources in water distribution systems. Compared with flow meters, pressure sensors are generally easier to install, require lower investment, and can be deployed more extensively throughout the network. Therefore, pressure-based methods have attracted increasing attention for leakage identification and localization. The underlying principle is that pipe leakage disturbs the hydraulic equilibrium of the WDN, typically resulting in pressure reductions and deviations from normal operating patterns. By analyzing the discrepancies between real-time pressure observations and expected hydraulic conditions, leakage events can be detected, and their potential locations can be inferred [26]. For example, Farley et al. [27] utilized hydraulic model simulations to optimize pressure sensor locations by quantifying the sensitivity of different monitoring points to potential burst events, enabling effective burst localization with a limited number of pressure instruments. Duan et al. [28] reviewed transient-based methods for water distribution system diagnosis, where transient pressure

responses generated by hydraulic disturbances were analyzed to identify leakage and other pipeline defects.

In recent years, ensemble learning has gained increasing attention in machine learning applications due to its ability to improve predictive performance and robustness by combining multiple base learners. In the field of water distribution systems, ensemble learning approaches have been increasingly explored for leakage detection and localization by integrating heterogeneous models and capturing complex relationships between hydraulic responses and leakage characteristics. For example, Zhou, et al. [29] developed an ensemble transfer learning one-dimensional convolutional neural network (TL1DCNN) for pipeline leak detection and localization by integrating multiple CNN-based base learners, achieving improved localization performance compared with individual models. Farghadan, et al. [30] proposed an ensemble Bayesian classifier framework for leak localization in water distribution networks by integrating multiple classifiers to identify leak locations under sensor limitations and model uncertainties. Ravichandran, et al. [31] proposed a gradient boosting tree-based ensemble learning approach for acoustic leak detection, which improved detection accuracy and robustness compared with conventional machine learning models. Fuentes and Pedrasa [32] developed an ensemble machine learning framework for WDN leak detection, demonstrating the potential of combining multiple ML models to improve leakage identification performance.

Although significant progress has been achieved in leakage detection and localization research, several challenges remain to be addressed. First, existing studies have demonstrated promising performance in leakage localization; however, comparatively limited attention has been devoted to robust multi-region localization classification in WDNs with complex topologies, particularly under operational uncertainties. Second, although various machine learning techniques have been increasingly applied to leakage detection and localization, the potential of ensemble learning to integrate the complementary strengths of different algorithms and improve localization robustness has received comparatively limited attention. In addition, the interpretability of data-driven leakage localization models remains an important consideration, as water utilities require not only accurate localization results but also an understanding of the monitoring information and hydraulic patterns underlying model decisions. Finally, existing studies have explored spatial, topological, and hydraulic information for leakage analysis from different perspectives, but a systematic framework that explicitly couples spatial/topological properties with hydraulic response characteristics for network zoning and subsequent localization classification remains relatively underexplored. These observations motivate the development of an interpretable ensemble learning framework that integrates spatial–hydraulic information for robust multi-region leakage localization in WDNs. This study proposes the following objectives:

1. To construct a WDN partitioning model that deeply couples spatial and hydraulic features, utilizing spatial distance and hydraulic similarity indicators to achieve rational network partitioning and provide physical support for accurate leakage localization.
2. To develop and compare multiple localization frameworks based on different machine learning models and ensemble strategies, systematically evaluating the differences in localization accuracy and robustness among Random Forest (RF), XGBoost, Multilayer Perceptron (MLP), ensemble learning approaches including Voting and Stacking constructed from these base models, and an independently trained LightGBM model.
3. To introduce the SHAP (SHapley Additive exPlanations) analysis method for multidimensional model interpretation and combine it with confusion matrix analysis to reveal the intrinsic relationships between pressure residuals, leakage regions, and model decision-making behavior from both global and local perspectives.

4. To evaluate the robustness and noise tolerance of the proposed framework, and to assess the stability of SHAP feature importance under varying levels of measurement uncertainty, thereby confirming its reliability and interpretability for practical leakage localization.

By achieving these objectives, this study aims to bridge the gap between the predictive capability of machine learning and the interpretability requirements of the water utility industry. Through accurate localization prediction and transparent decision analysis, the proposed framework is expected to provide technical support for the efficient operation and maintenance of WDNs.

2. Research Methodology

To construct a high-accuracy, robust, and interpretable leakage localization framework for urban water distribution networks (WDNs), this study proposes an integrated methodology combining spatial–hydraulic regional partitioning, stochastic scenario generation, heterogeneous model ensembling, and multi-dimensional interpretability analysis. As illustrated in Figure 1, the overall workflow systematically progresses through four interconnected stages: first, coupling spatial topological distances and hydraulic head similarities to divide the network into physically meaningful operational zones; second, simulating diverse leakage scenarios via Monte Carlo sampling on virtual mid-pipe nodes during nighttime low-demand conditions and applying Pearson correlation alongside ANOVA feature selection to identify an optimal, non-redundant sensor subset; third, benchmarking heterogeneous base learners (RF, XGBoost, and MLP) against an independent LightGBM baseline, and fusing their complementary mechanisms using Voting and Stacking ensemble architectures (with a Logistic Regression meta-learner); and finally, evaluating localization accuracy and robustness under measurement noise perturbations while employing KernelExplainer SHAP analysis and confusion matrix cross-examination to uncover the underlying hydraulic logic governing model predictions.

2.1. Pipeline Leakage Simulation

To simulate potential leakage locations and their hydraulic responses in WDNs, this study employed the virtual node method by introducing virtual nodes at the midpoints of pipelines to represent possible leak points. This approach can effectively capture pressure fluctuations and flow redistribution caused by leakage without altering the original network topology, thereby improving numerical stability while maintaining computational efficiency. Previous studies have demonstrated the applicability of this method in WDN analysis. For example, Lee and Choi [33], as well as Kumar et al. [34], modeled hydraulic behaviors in pipeline networks based on virtual nodes. In the present study, emitters were assigned to the virtual nodes to represent potential leakage points. The emitter-based leakage model establishes a direct relationship between the leakage flow rate and the pressure at the corresponding virtual node. Accordingly, the leakage flow rate is calculated as:

$$Q = CP^{0.5} \tag{1}$$

where Q denotes the leakage flow rate, P represents the pressure at the leakage node, and C is the emitter coefficient.

According to the EPANET 2.2 emitter formulation, the emitter coefficient is a scenario-dependent parameter rather than a universal constant. Previous studies have adopted different strategies for assigning this coefficient according to their simulation objectives. For example, Ajoodani et al. [35] assumed a constant emitter coefficient for simplified leakage simulations, whereas Amoatey et al. [36] generated leakage scenarios by randomly

varying emitter coefficients within a predefined range. Following these practices, the emitter coefficient was treated as a variable parameter in this study and uniformly sampled within the interval [0.1, 0.8] to represent leakage events with different severities. This range was selected to generate diverse leakage scenarios while maintaining physically meaningful and numerically stable hydraulic simulations. In this study, burst leakage and progressive leakage were not explicitly modeled as separate leakage mechanisms. Instead, all leakage events were generated using the same scenario generation framework, with leakage severity represented by variations in the emitter coefficient. This simplification was adopted to evaluate the effectiveness of the proposed monitoring point optimization method while avoiding additional assumptions regarding leakage evolution.

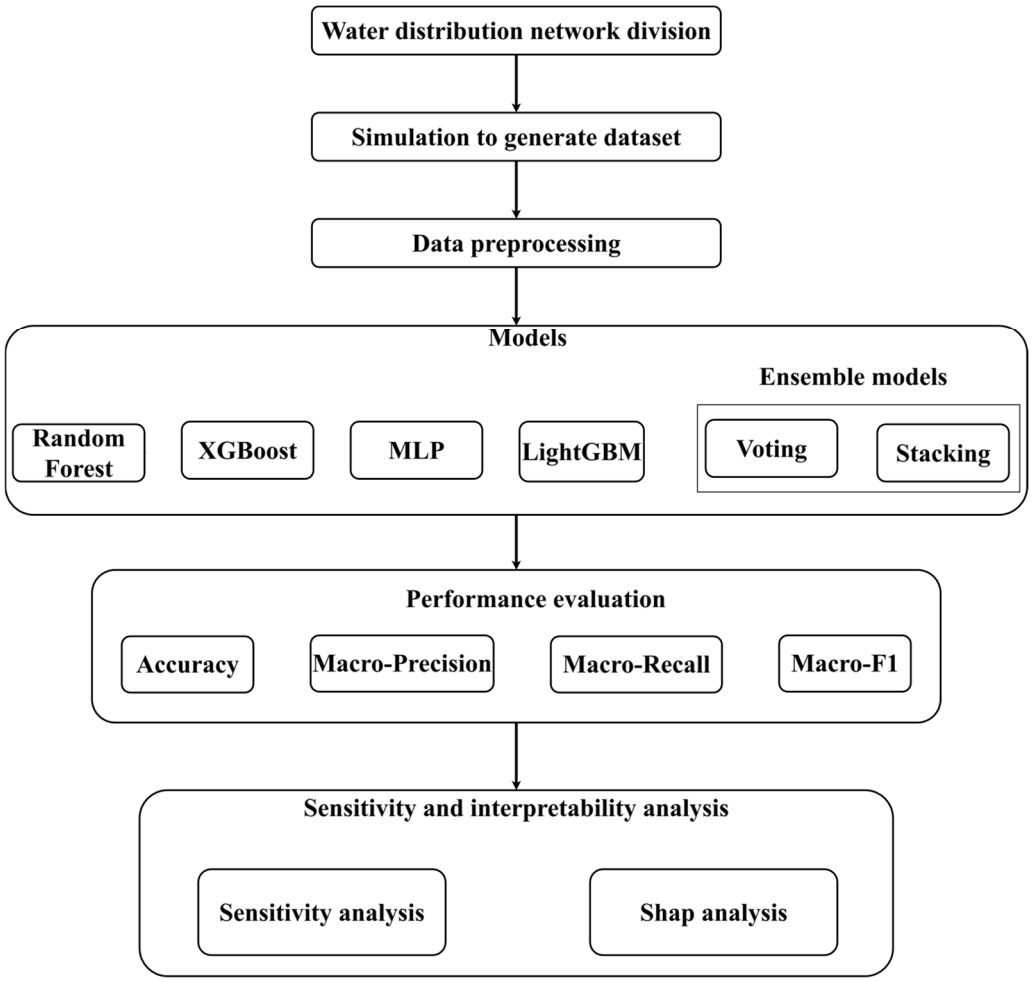


Figure 1. Framework of the study.

To highlight the local disturbances caused by leakage, this paper uses the pressure residual as a characteristic quantity, which helps enhance the model's sensitivity to leakage-induced hydraulic disturbances. It is defined as:

$$PR_i = P_i^N - P_i^L \tag{2}$$

where PR_i denotes the pressure residual at node i, and P_i^N is the pressure under normal operating conditions, and P_i^L is the pressure under leakage conditions.

2.2. Water Distribution Network Division

To achieve a reasonable partitioning of the pipeline network structure and support subsequent leakage localization analysis, this study proposes a regional partitioning method in-

tegrating spatial information and hydraulic characteristics. The overall framework consists of two stages: initial spatial partition construction and regional assignment determination of virtual nodes.

First, based on the spatial coordinates of monitoring nodes, the K-means clustering algorithm was employed to divide the nodes into K initial regions. This process considers only spatial location relationships, aiming to establish the fundamental partition structure of the network and provide regional centroid references for subsequent analysis.

On this basis, to simultaneously account for spatial proximity and hydraulic consistency, a coupled assignment index was introduced to determine the regional affiliation of virtual nodes. For a virtual node j, the assignment index is defined as follows:

$$S_k(j) = \frac{1}{\left|h_j - \overline{h}_k\right| + \epsilon} \cdot \exp\left(-\beta \frac{\left\|x_j \mu_k\right\|}{d_{\max}}\right) \tag{3}$$

where x_j denotes the spatial coordinates of virtual node j, and h_j represents its hydraulic head; μ_k is the spatial centroid of the cluster k, and $\overline{h}_k$ is the average hydraulic head of the cluster; β is the spatial constraint strength parameter, which was set to 5.0 in this study based on preliminary experiments. This value was selected because it provided a suitable balance between spatial proximity and hydraulic similarity, avoiding excessive spatial clustering fragmentation while maintaining meaningful hydraulic patterns; $d_{\max}$ is the maximum spatial scale of the WDN used for distance normalization; and ϵ is a small positive constant introduced to avoid division by zero, which was set to 0.01 in this study, as this value has a negligible effect on the computed results.

Ultimately, the region to which a virtual node belongs is determined by the following criteria:

$$k^* = \text{argmax}_k S_k(j) \tag{4}$$

This method integrates geometric structure and hydraulic characteristics in the regional partitioning process by coupling spatial distance with hydraulic head differences, thereby improving the physical rationality of the partitioning results and enhancing the accuracy of subsequent analyses.

2.3. Prediction Model

This study selected RF, XGBoost, and MLP as representative base learners considering their complementary modeling characteristics and widespread adoption in infrastructure-related prediction tasks. RF was included as a bagging-based tree ensemble that provides robustness against noise and feature variability, while XGBoost was selected as a boosting-based model with strong capability in capturing complex nonlinear relationships and feature interactions. MLP was introduced as a neural network-based learner to represent nonlinear patterns through hierarchical feature transformations. These models were selected to provide diverse learning mechanisms, including tree-based bagging, gradient boosting, and neural representation learning, thereby enabling a comprehensive evaluation of different modeling paradigms. In addition, LightGBM was included as an independent benchmark model owing to its efficient gradient boosting framework and strong predictive performance, providing an additional reference for performance comparison with the proposed ensemble approaches. The ensemble learning methods Voting and Stacking were then constructed using RF, XGBoost, and MLP as heterogeneous base learners to further investigate whether integrating complementary learning mechanisms could improve prediction stability and overall classification performance. Based on these models, a progressive comparison framework ranging from individual learners to ensemble approaches

was constructed to systematically evaluate the performance differences among various modeling strategies in the classification task.

2.3.1. Random Forest

RF is an ensemble learning method that constructs a strong learner by combining multiple decision trees. This method employs bootstrap sampling to generate multiple training subsets and introduces a random feature selection mechanism during the tree construction process, thereby effectively reducing the correlation among individual models. By aggregating the prediction results of all decision trees, RF can improve model robustness and generalization ability while maintaining high prediction accuracy [37].

2.3.2. Gradient Boosting Decision Tree (GBDT)

Gradient boosting decision tree (GBDT) models build an ensemble of decision trees sequentially, with each newly added tree trained to correct the residual errors of the existing ensemble. In this study, XGBoost and LightGBM, two representative GBDT-based algorithms, were adopted. Although both algorithms follow the same additive boosting framework, XGBoost improves predictive performance through regularized objective optimization and exact or approximate split finding, whereas LightGBM emphasizes computational efficiency by employing histogram-based feature discretization together with leaf-wise tree growth. Consequently, the final prediction is generated by aggregating the outputs of multiple decision trees [38,39]:

$$\widehat{y}_{i,\mathrm{GBDT}} = \sum_{k=1}^{K} f_k(x_i) \tag{5}$$

where x_i denotes the input feature vector of the i-th sample, K represents the number of decision trees, and $f_k(\cdot)$ denotes the prediction function of the k-th tree. During training, each tree is sequentially optimized to minimize the overall objective function.

2.3.3. Multilayer Perceptron (MLP)

MLP is a typical feedforward neural network architecture consisting of an input layer, one or more hidden layers, and an output layer. Information is transmitted between layers through fully connected connections, while nonlinear activation functions are introduced in the hidden layers, enabling the model to capture complex nonlinear relationships. MLP learns model parameters through the backpropagation algorithm and has been widely applied in various supervised learning tasks.

2.3.4. Voting

The Voting classifier is an ensemble learning method that combines the prediction results of multiple different base models to obtain the final classification result. Its core idea is similar to the principle of "majority voting." In hard voting, the class predicted by the majority of models is selected as the final output, whereas in soft voting, the predicted probabilities from different models are weighted and averaged before determining the final class. Given that soft voting considers the confidence levels of individual base models, this study employed the soft voting mechanism to achieve finer-grained decision integration. By integrating the strengths of multiple models, the Voting classifier can generally improve overall generalization ability and stability while reducing the bias or overfitting problems associated with a single model [40]. Figure 2 illustrates the schematic framework of this algorithm [41].

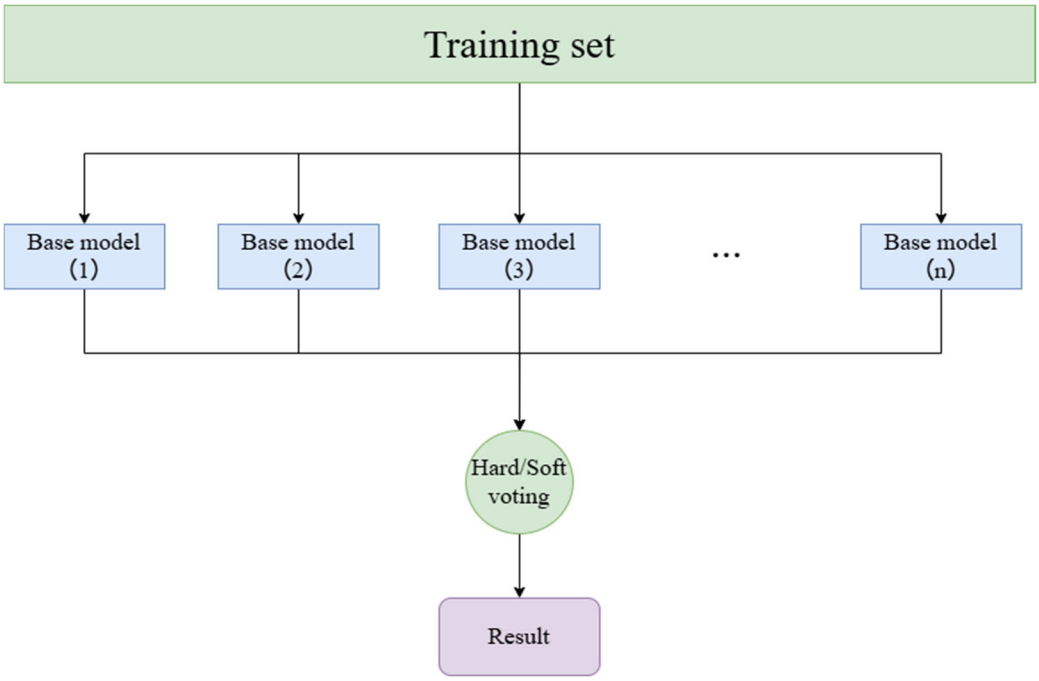


Figure 2. Framework of voting.

2.3.5. Stacking

Stacking is a hierarchical ensemble learning method whose fundamental idea is to improve overall predictive performance by constructing a multi-level model structure. In this method, multiple base models are first trained independently on the original training data to generate corresponding prediction outputs. Subsequently, the prediction results of the base models are used as new input features for a meta-model, which performs further learning to optimize the final prediction results. Compared with single models, Stacking can effectively integrate the representational capabilities of different models, reduce generalization error, and improve prediction stability and accuracy. Therefore, it has strong application value in complex tasks. Figure 3 presents a schematic illustration of this algorithm [42].

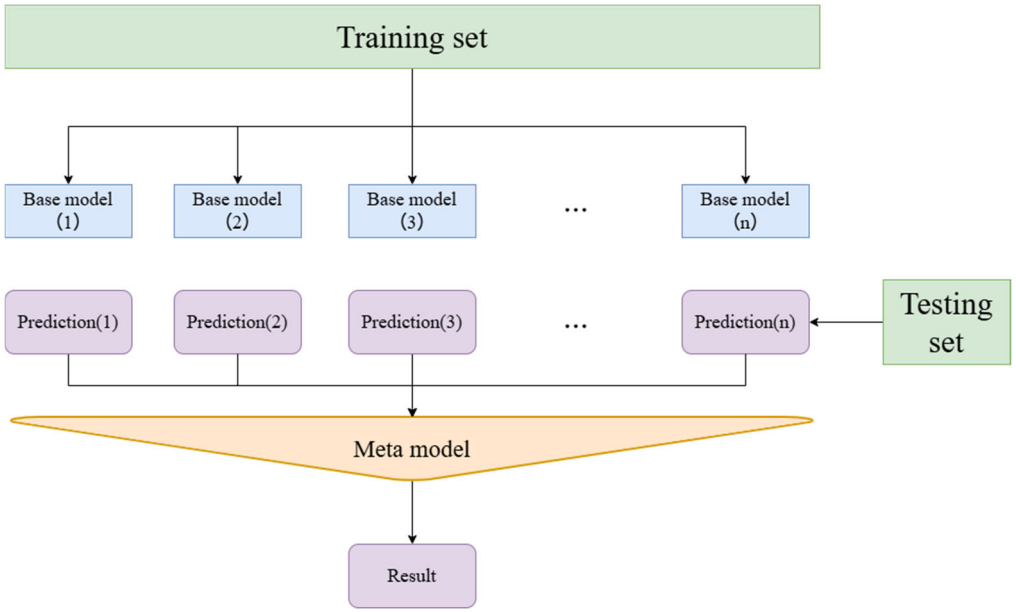


Figure 3. Framework of stacking.

2.4. Model Performance Metrics

This study adopted Accuracy, Precision, Recall, and F1-score as evaluation metrics. For multi-classification tasks, these metrics were calculated based on the confusion matrix using a one-versus-rest strategy. These metrics were selected because they provide complementary perspectives on model performance. Accuracy reflects the overall classification correctness, while Precision and Recall evaluate the model's ability to correctly identify samples of each class from different perspectives. The F1-score integrates Precision and Recall, providing a balanced measure of classification performance, particularly when class distributions are imbalanced. To account for the multi-class nature of the leakage zone classification task, macro-averaged metrics were employed by assigning equal importance to each class. The evaluation metrics were defined as follows:

$$Accuracy = \frac{\sum_{i=1}^{K} TP_i}{N} \tag{6}$$

$$Macro\text{-}Precision = \frac{1}{K}\sum_{i=1}^{K} \frac{TP_i}{TP_i + FP_i} \tag{7}$$

$$Macro\text{-}Recall = \frac{1}{K}\sum_{i=1}^{K} \frac{TP_i}{TP_i + FN_i} \tag{8}$$

$$Macro\text{-}F1 = \frac{1}{K}\sum_{i=1}^{K} \frac{2TP_i}{2TP_i + FP_i + FN_i} \tag{9}$$

where K represents the number of classes, N denotes the total number of samples, TP_i represents the correctly classified samples of class i, FP_i represents the samples from other classes incorrectly assigned to class i, and FN_i represents the samples of class i incorrectly classified into other classes.

2.5. SHAP

SHAP is a model interpretation method based on Shapley values from cooperative game theory, which is used to quantify the contribution of each feature to model prediction results. In this method, each feature is regarded as a "player," and its average influence on the final prediction is obtained by calculating its marginal contribution across all possible feature combinations. In this way, SHAP enables both global and local interpretation of complex models. SHAP satisfies theoretical properties such as local accuracy, missingness, and consistency, allowing it to provide intuitive interpretability analysis for black-box models while maintaining explanatory reliability. Consequently, it has been widely applied in the field of machine learning model interpretation. Among the commonly used SHAP methods, TreeSHAP is suitable for tree-based models, such as random forests and gradient boosting models, where it can efficiently compute exact SHAP values. DeepSHAP is commonly applied to deep neural networks by combining Shapley values with deep learning-based approximation strategies. KernelSHAP (KernelExplainer) is a model-agnostic method that estimates SHAP values through a weighted linear regression approach and can therefore be applied to virtually any machine learning model, although with relatively higher computational cost. The SHAP value is calculated based on the Shapley value from cooperative game theory as follows [43]:

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left[f_{S\cup\{i\}}\left(x_{S\cup\{i\}}\right) - f_S(x_S) \right] \tag{10}$$

where $\phi_i(fx)$ denotes the SHAP value of feature i, F is the set of all input features, S is any subset of features excluding feature i, and $f_S(x_S)$ denotes the model prediction based

on the feature subset S. The difference $f_{S\cup\{i\}}\left(x_{S\cup\{i\}}\right) - f_S(x_S)$ represents the marginal contribution of feature i, and the weighted average over all possible subsets yields the SHAP value.

3. Results

3.1. Data Description

This study employed the L-Town water distribution model from the BattLeDIM competition [44]. L-Town is a small hypothetical urban WDN consisting of 785 nodes, 909 pipes, one pump, and three valves. In addition, 33 pressure sensors are installed at different nodes within the L-Town network. The L-Town network is adopted as the evaluation case study owing to its established status as an authoritative benchmark from the BattLeDIM competition, which ensures the reproducibility and comparability of our findings against state-of-the-art methodologies. Furthermore, despite its moderate scale, the network features highly heterogeneous topological attributes and integrates critical operational components (such as the pump and valves) capable of replicating complex, realistic hydraulic dynamics. Crucially, the deployment of 33 pressure sensors against 785 nodes represents a highly sparse sensing configuration. This alignment with the practical budgetary and infrastructural constraints of real-world water utilities provides a rigorous environment to validate the robustness of the proposed leakage localization framework.

Figure 4 illustrates the topology of the L-Town network and the locations of the pressure sensors. In the figure, pressure sensors are marked in red, while the green squares represent reservoirs. Figure 5 presents the L-Town network after the introduction of virtual nodes.

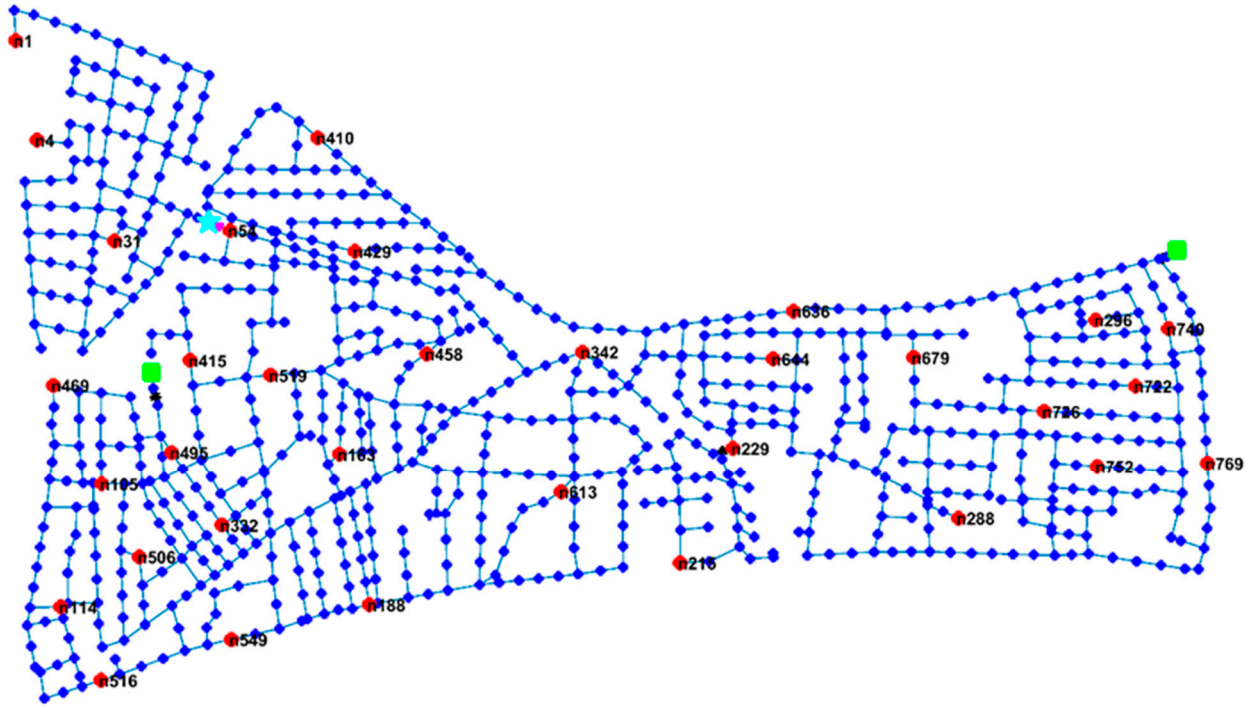


Figure 4. L-Town network topology (red: pressure sensors; green squares: reservoirs).

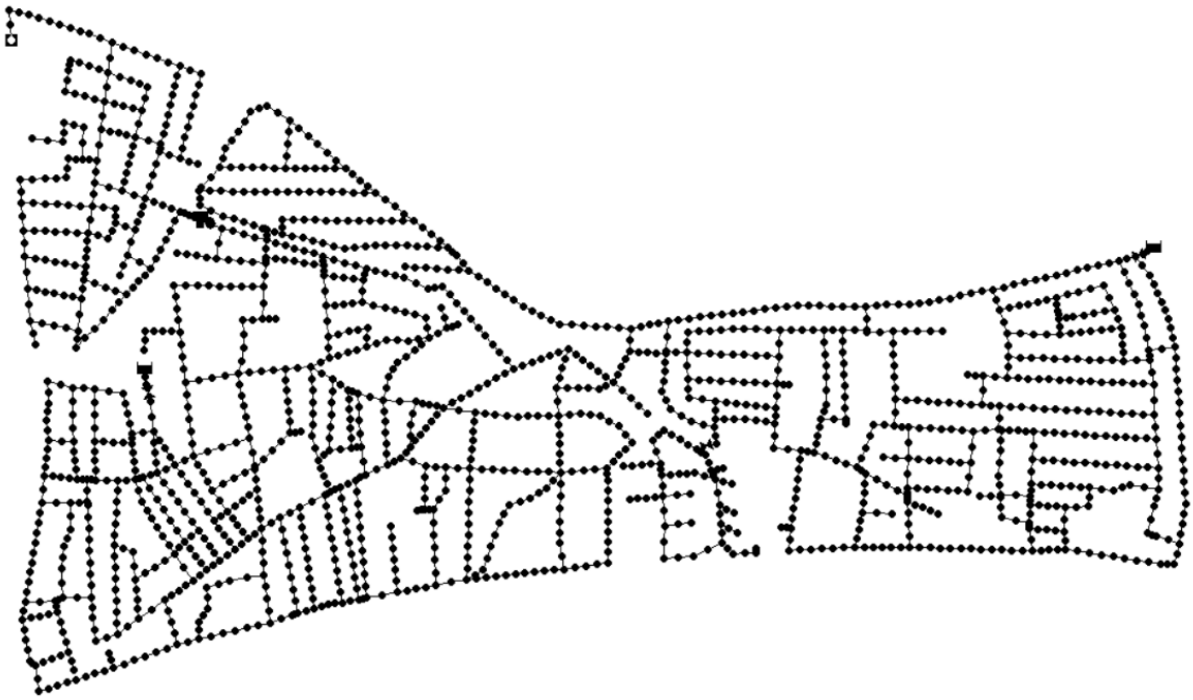


Figure 5. L-Town network with virtual nodes added.

In this study, the L-Town network was divided into seven regions. The number of partitions was determined with reference to the scale of the network and the number of monitoring points, aiming to ensure sufficient sample size within each region while balancing computational complexity and spatial resolution. Figure 6 presents the virtual node partitioning results of the L-Town network.



Figure 6. Virtual node partitioning of the L-Town network (triangles: monitoring points).

This study constructed a WDN simulation environment based on WNTR and performed hydraulic calculations using EPANET [45,46]. Simulation analysis was conducted under nighttime low-demand conditions with a total duration of 4 h, where the total baseline network user demand at the final hour was 171.3 m^3/h (a significant reduction compared to the network's total base demand of 634.72 m^3/h to reflect midnight hydraulics). The nodal pressures at the final simulation time (the 4th hour) were extracted as steady-state responses to reduce the influence of transient fluctuations on the results. First, the baseline pressure distribution under leak-free conditions was obtained to represent the normal operating state of the system.

Monte Carlo simulation is a widely adopted stochastic sampling approach for representing uncertainty and exploring possible system states by repeatedly generating random samples from predefined probability distributions [47]. On this basis, the Monte Carlo method was adopted to construct leakage scenarios to improve sample diversity and spatial coverage. In each independent simulation scenario, 1–5 nodes were randomly selected as leakage locations within only one specific partition, meaning that concurrent leakages across multiple partitions were not considered in this multi-class classification setup. To simulate varying leakage intensities, the emitter coefficient for each selected node was randomly generated via a uniform distribution ranging from 0.1 to 0.8. Under the nighttime pressure regime, this resulted in an individual node leakage flow range of approximately 6 m^3/h to 65 m^3/h, driving the total leakage flow of a single scenario to vary from a minor incident to a severe burst relative to the low nighttime background demand.

For the seven partitions obtained from the network division, 150 leakage samples were generated for each partition, resulting in a dataset containing 1050 samples in total. In this dataset, the pressure residual vector of the monitoring nodes was used as the input feature, while the partition ID corresponding to the localized leakage location was used as the output target. The model's objective is to identify the specific leaking partition without prior knowledge of the exact number of active leaks, their precise node IDs, or the total leakage flow.

During the data preprocessing stage, stratified sampling was first employed to divide the dataset into training and testing sets at a ratio of 8:2. Specifically, 840 samples were allocated to the training set, and 210 samples were reserved as the independent testing set. This strategy ensured consistent class distributions in both subsets and reduced the influence of class imbalance on model evaluation. Subsequently, the input features were standardized to eliminate the effects of dimensional differences on model training and to improve convergence speed and model stability.

Initially, a total of 33 candidate monitoring nodes distributed across the network were considered as features. For feature selection, Pearson correlation coefficients were first used to quantify the linear correlations among these 33 candidate features, thereby identifying highly correlated feature pairs. In this study, feature pairs with an absolute Pearson correlation coefficient greater than 0.9 ($|r| > 0.9$) were considered highly correlated. For these highly correlated features, one-way analysis of variance (ANOVA) was further applied to evaluate their classification discrimination ability through F-values. A larger F-value indicates stronger discriminative capability among different classes (partitions).

To avoid data leakage, the independent testing set was excluded from all preprocessing procedures involving statistical estimation. Feature selection based on Pearson correlation analysis and ANOVA F-values was performed solely on the training set, and the resulting feature subset was then applied to the testing set. Likewise, feature standardization parameters (mean and standard deviation) were estimated from the training set and subsequently used to transform both the training and testing sets. Therefore, no information from the

testing set was involved during model development, ensuring an unbiased evaluation of model performance.

Based on this analysis, redundant features were systematically removed. For highly correlated feature pairs, the feature with the higher F-value and greater classification contribution was preferentially retained. Furthermore, the geographic and spatial distribution of the nodes across the 7 partitions was carefully cross-verified during the elimination process to avoid spatial clustering of sensors and ensure comprehensive monitoring coverage over the entire network topology. Consequently, the feature dimension was successfully reduced from the initial 33 candidate points to an optimal subset of 8 monitoring points. This process successfully eliminates feature redundancy while preserving critical information as much as possible, thereby improving both the generalization ability and computational efficiency of the model. Figure 7 presents the correlation matrix heatmap of these 8 final retained monitoring points, which serve as the definitive input features for the leak localization models.

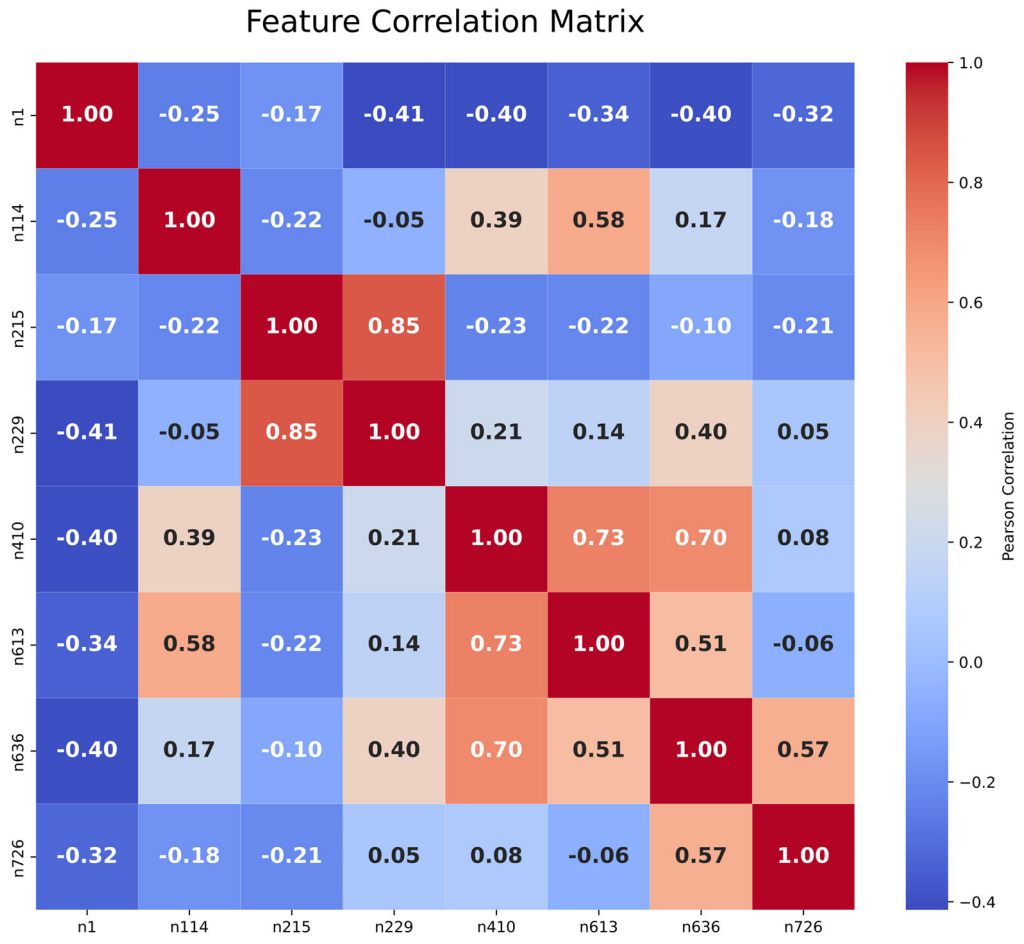


Figure 7. Correlation matrix heatmap.

3.2. Model Evaluation

Hyperparameter optimization was performed for all individual models. Grid search was not adopted because its exhaustive evaluation of all parameter combinations would result in high computational costs. Therefore, Random Search and Optuna were compared in preliminary experiments. As shown in Figure 8 and Table 1, both methods achieved comparable macro-F1 scores, while Random Search showed a time advantage. Thus, Random Search was selected for the final hyperparameter optimization. For each model, 100 randomly sampled parameter combinations were evaluated using five-fold cross-

validation, with the mean macro-F1 score as the optimization criterion. The optimized hyperparameters are summarized in Table 2.

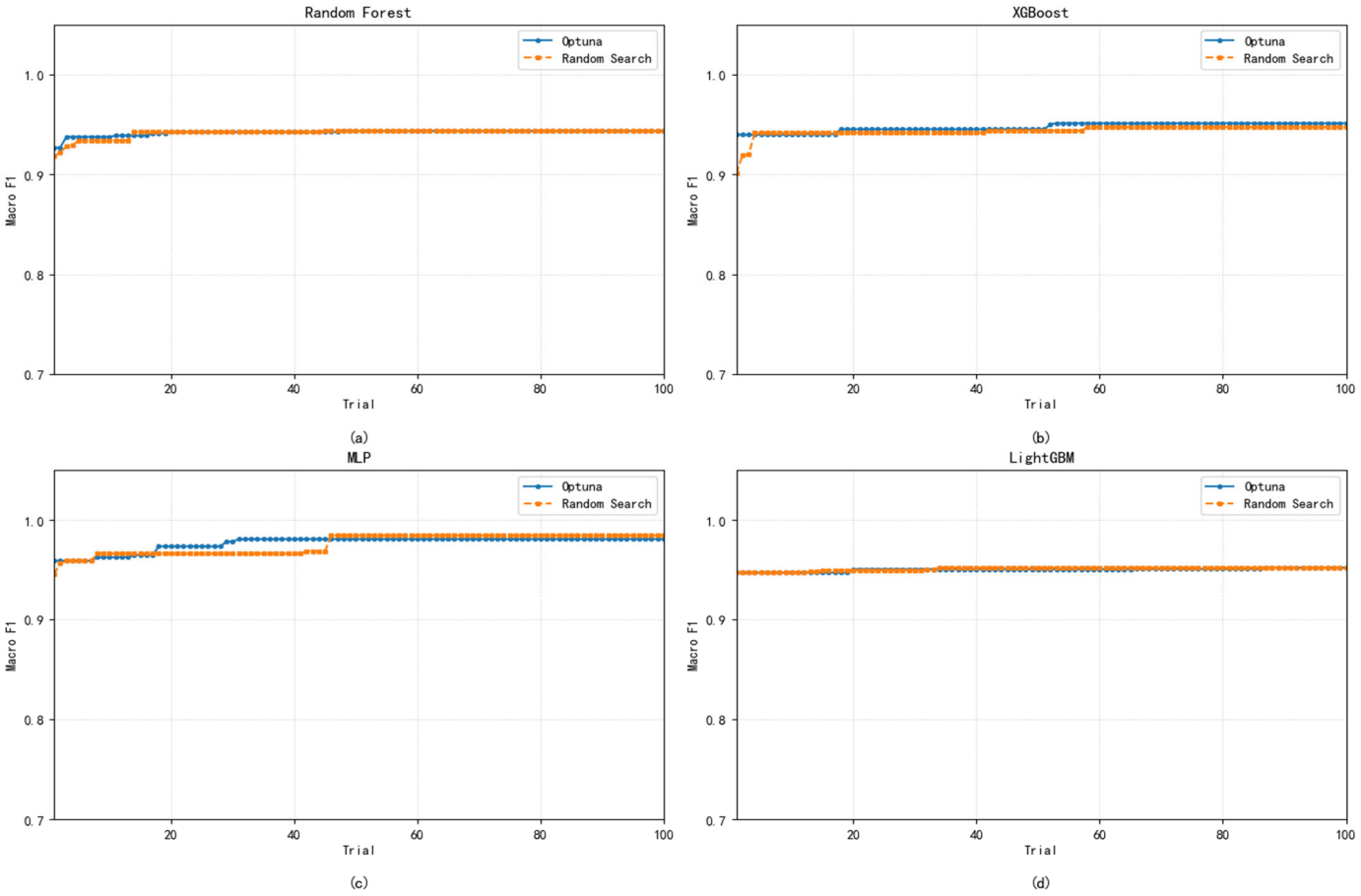


Figure 8. Performance comparison of Random Search and Optuna: (**a**) Random Forest, (**b**) XGBoost, (**c**) MLP, and (**d**) LightGBM.

Table 1. Hyperparameter optimization time.

Model	Random Search	Optuna
RF	232.78 s	573.67 s
XGBoost	129.23 s	613.60 s
MLP	124.25 s	865.19 s
LightGBM	608.66 s	323.75 s

For the Stacking ensemble model, RF, XGBoost, and MLP were selected as heterogeneous base-learners due to their complementary learning behaviors, while Logistic Regression was employed as the second-stage meta-learner. An out-of-fold (OOF) prediction strategy based on 5-fold cross-validation was strictly adopted to generate meta-features for training the meta-learner, thereby preventing data snooping and information leakage. In addition, a heterogeneous Voting ensemble model was developed using the same three optimized base learners through a soft-voting mechanism that averages the predicted class probabilities. To further provide a comprehensive benchmark, an optimized LightGBM model was independently implemented as an additional gradient boosting-based comparison model. The performance of all optimized individual models (RF, XGBoost, MLP, and LightGBM) and ensemble architectures (Stacking and Voting) was finally evaluated on an independent test set. The detailed performance metrics are summarized in Table 3, and the quantitative comparisons among different models are illustrated in Figure 9.

Table 2. Hyperparameter configurations.

Model	Hyperparameter	Search Space	Value
RF	n_estimators	[100, 600], step = 25	275
	max depth	[3, 30]	16
	min_samples_split	[2, 20]	5
	min_samples_leaf	[1, 10]	1
	max_features	{"sqrt", "log2"}	sqrt
XGBoost	n_estimators	[100, 800], step = 25	750
	learning_rate	[0.01, 0.2], log = True	0.0837
	max_depth	3, 12	11
	subsample	0.6, 1.0	0.8824
	colsample_bytree	0.6, 1.0	0.7912
MLP	hidden_layer_sizes	[32, 256], step = 16	(176, 96)
	activation	{"relu", "tanh"}	relu
	alpha	$[1 \times 10^{-5}, 1 \times 10^{-2}]$, log = True	0.00001
	learning_rate_init	$[1 \times 10^{-4}, 5 \times 10^{-2}]$, log = True	0.00109
LightGBM	n_estimators	[100, 800], step = 25	250
	learning_rate	[0.01, 0.2], log = True	0.1081
	num_leaves	[15, 150]	108
	max_depth	[3, 15]	4

Table 3. Performance metrics for models.

Model	Accuracy	Macro-Precision	Macro-Recall	Macro-F1
RF	0.9524	0.9533	0.9524	0.9522
XGBoost	0.9476	0.9494	0.9476	0.9479
MLP	0.9524	0.9532	0.9524	0.9523
LightGBM	0.9619	0.9628	0.9619	0.9617
Voting	0.9571	0.9579	0.9571	0.9570
Stacking	0.9667	0.9671	0.9667	0.9666

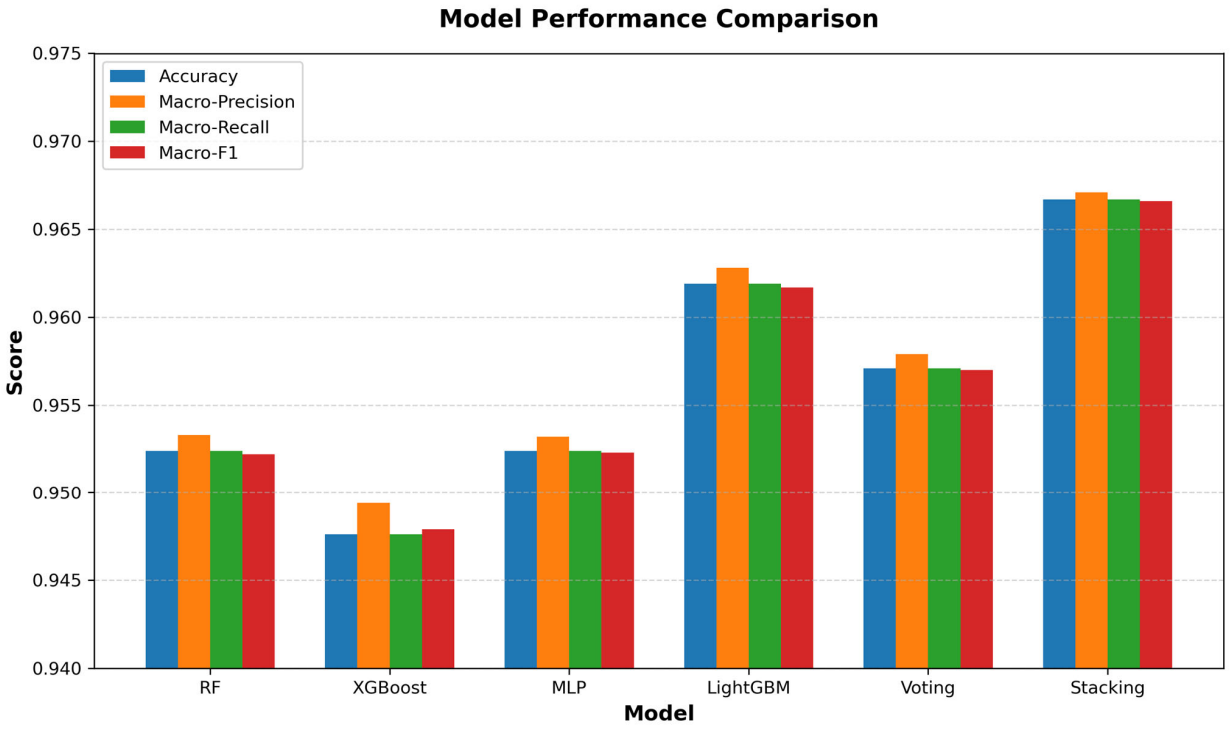


Figure 9. Model performance comparison.

The performance comparison of different models is presented in Table 2. All models achieved high classification accuracy, with values exceeding 94%, indicating that the pressure residual features contained sufficient spatial information for distinguishing different leakage regions. Among the individual models, RF and MLP showed comparable performance, achieving accuracies of 0.9524 and 0.9524, respectively, while XGBoost obtained a slightly lower accuracy of 0.9476. These results demonstrate that both tree-based and neural network models were capable of capturing the nonlinear relationships between pressure residual patterns and leakage locations. The consistent performance among individual models also indicates the effectiveness of pressure residuals as discriminative features for leakage region classification.

The ensemble models further enhanced the classification performance. The Voting model achieved an accuracy of 0.9571, while the Stacking model obtained the highest performance, with an accuracy of 0.9667 and a macro F1-score of 0.9666. Compared with the individual learners, the Stacking framework benefited from integrating the complementary predictive information of heterogeneous base models, leading to improved classification capability.

To further validate the overall superiority of the proposed models, a Copeland ranking analysis was conducted by comprehensively considering multiple evaluation metrics. Unlike comparisons based on a single performance indicator, the Copeland method aggregates pairwise comparisons across all metrics to provide a more robust overall ranking [48]. Figure 10 shows the results of Copeland. Although the Stacking model showed only marginal improvements in individual metrics, it achieved the highest overall ranking. This result suggests that the Stacking framework can provide a more balanced classification performance across different evaluation criteria.

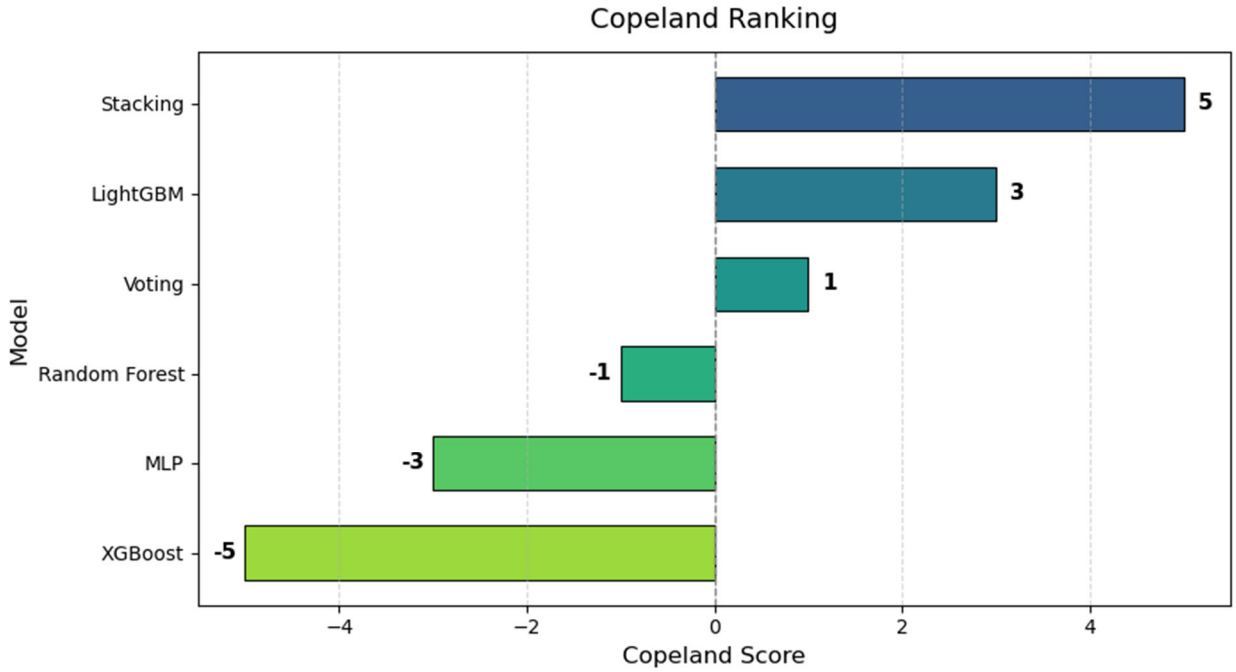


Figure 10. Copeland ranking.

3.3. Model Interpretability

To evaluate the operational reliability of the developed Stacking ensemble model and unpack its "black-box" decision-making process, a joint diagnosis was conducted by cross-examining the localized classification errors from the confusion matrix (Figure 11), the SHAP feature importance profiles (Figure 12), and the virtual node partitioning of the L-Town network (Figure 6). Specifically, KernelExplainer was employed to interpret the

final Stacking ensemble by treating the entire ensemble pipeline, including both the base learners and the meta-learner, as a unified black-box predictor, thereby quantifying the contribution of the original input features to the final classification decisions.

As illustrated in the confusion matrix, the Stacking framework achieves an accuracy of 100% for Classes 1, 4, 5, and 6, with all 30 samples in each class correctly localized. Specifically, Class 1 (Region 1) is characterized by a dominant contribution from monitoring node n1, which ranks as the top predictor by a substantial margin. Geographically, n1 is deployed at the far northwestern terminal of the network, providing a spatially distinctive monitoring location for Region 1. Similarly, the 100% accuracy achieved in Class 4 (Region 4) is associated with the dominant contribution of node n114, located in the southwestern part of the network. These results indicate that the identified monitoring nodes provide important discriminative information for the corresponding regions.

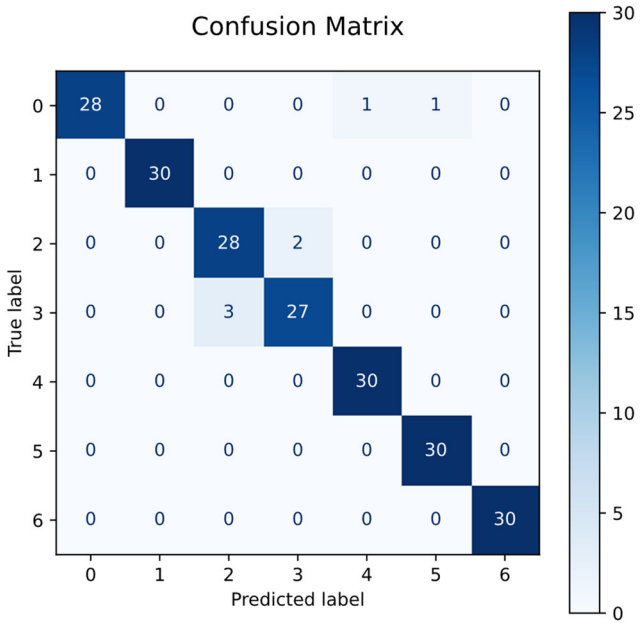
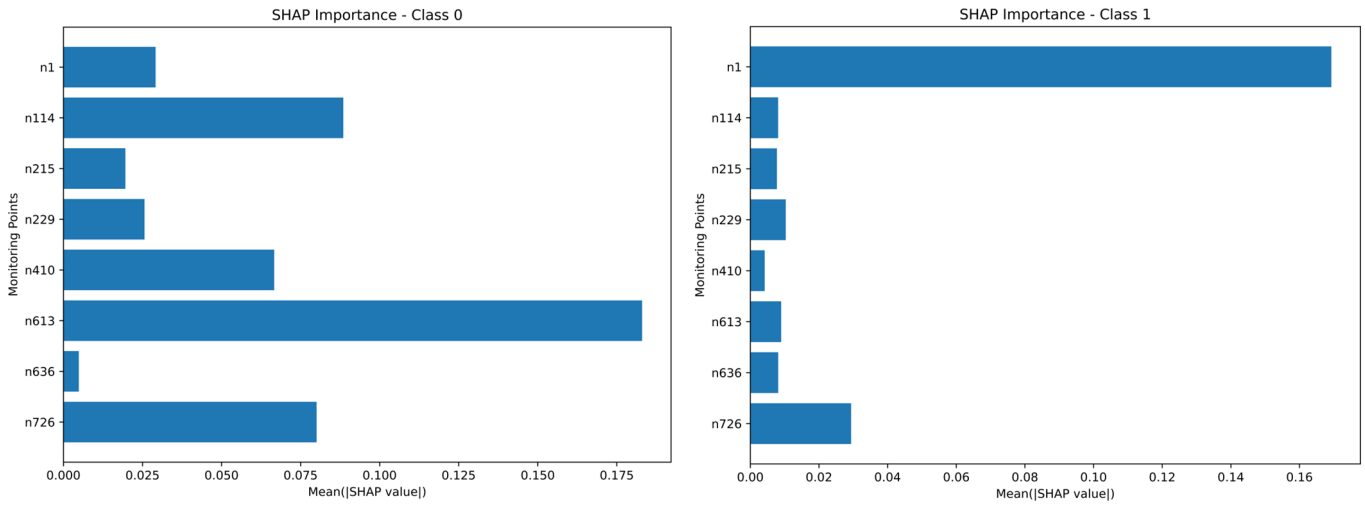


Figure 11. Confusion matrix.



(**a**) (**b**)

Figure 12. *Cont.*

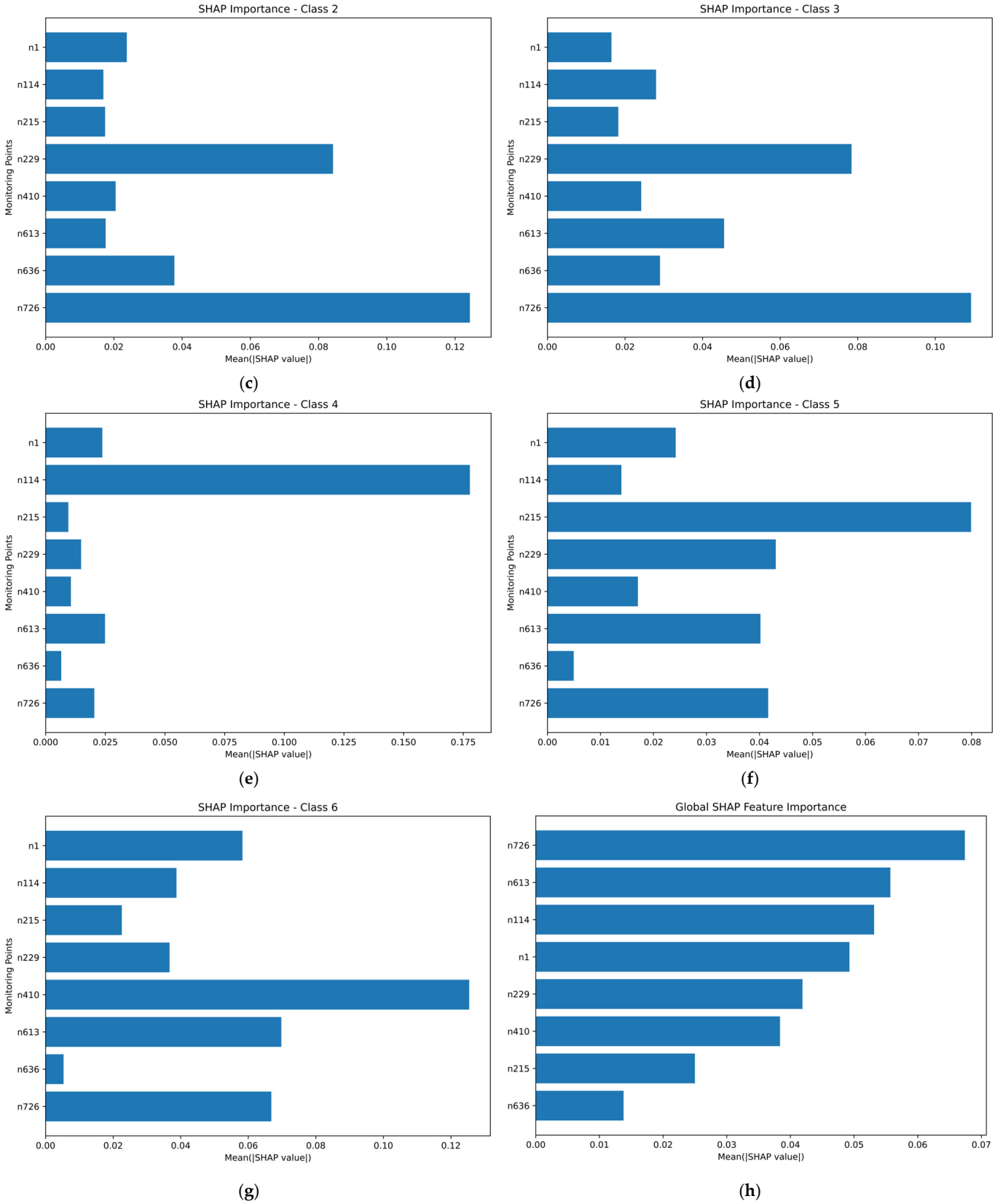


Figure 12. SHAP feature importance analysis: (**a**–**g**) class-specific SHAP feature importance results for the seven categories, and (**h**) the global SHAP feature importance.

The misclassifications observed in the confusion matrix are mainly concentrated between a small number of neighboring classes. The most prominent case is the bidirectional confusion between Class 2 and Class 3, where 2 instances of Class 2 are misclassified as Class 3 and 3 instances of Class 3 are misclassified as Class 2. The SHAP profiles show that node n726 is the dominant predictor for both classes. Region 2 and Region 3 are spatially contiguous, and the common importance of n726 indicates that this monitoring location provides substantial information for the classification of both regions. The bidirectional misclassification pattern is therefore consistent with a certain degree of similarity in the feature representations of these two regions. Despite this localized overlap, the majority of samples in both classes are correctly identified, indicating clear overall discrimination between the two regions.

A similar pattern is observed for Class 0 (Region 0), although with substantially fewer misclassifications. One sample from Class 0 is classified as Class 4, while another is classified as Class 5. The SHAP profile for Class 0 identifies node n613 as its dominant predictor. The occurrence of these isolated errors between neighboring regions suggests that a small number of samples exhibit feature characteristics similar to those of adjacent classes. These localized overlaps have a limited effect on the overall classification performance of the framework.

The global SHAP feature-importance profile summarizes the relative contribution of the monitoring nodes to the overall classification task. Node n726 exhibits the highest overall importance, followed by n613, n114, and n1. The high importance of these nodes indicates that they provide substantial contributions to the model predictions across the considered regions. In particular, the prominence of n726 and n613 in the global ranking is consistent with their strong contributions in the class-specific SHAP profiles, while n114 and n1 are also among the most influential features for the classification of their corresponding regions. Overall, the integrated confusion-matrix and SHAP analysis provide a consistent interpretation of the classification results, linking the observed regional performance with the relative contribution of individual monitoring nodes. The high classification accuracy across most regions, together with the localized nature of the observed errors, demonstrates the effectiveness of the proposed zoning and localization framework.

3.4. Sensitivity Analysis

To account for the discrepancy between simulated and real-world measurements and to evaluate the robustness of the proposed Stacking ensemble model, a sensor noise sensitivity analysis was performed by injecting different levels of noise into the simulated sensor data. The purpose of this analysis is to emulate the inevitable discrepancy between simulated and real-world measurements while simultaneously investigating the stability and robustness of the localization model under varying noise intensities. The localization accuracy under different noise levels is presented in Figure 13. Furthermore, the influence of noise perturbations on the contribution of monitoring points was analyzed using SHAP values, and the variations in feature importance under different noise conditions are shown in Figure 14.

Figure 13 illustrates the variation in localization accuracy of the proposed Stacking ensemble model across different levels of noise. As observed, the accuracy exhibits a gentle downward trend as noise intensity increases. Within the 0% to 10% noise range, the model maintains high accuracy with small performance degradation. Even under higher noise intensities up to 20%, the accuracy remains above the 0.85 threshold. These results suggest that the proposed model demonstrates reasonable resilience against measurement perturbations, indicating its potential applicability in practical environments.

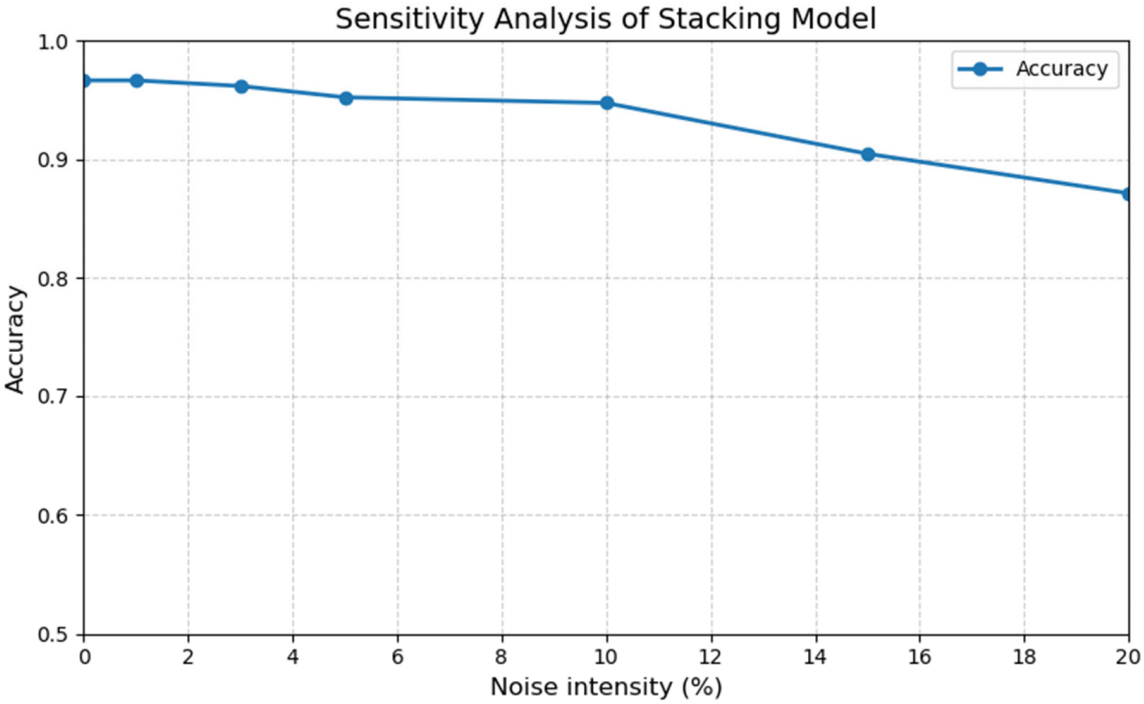


Figure 13. Model sensitivity analysis.

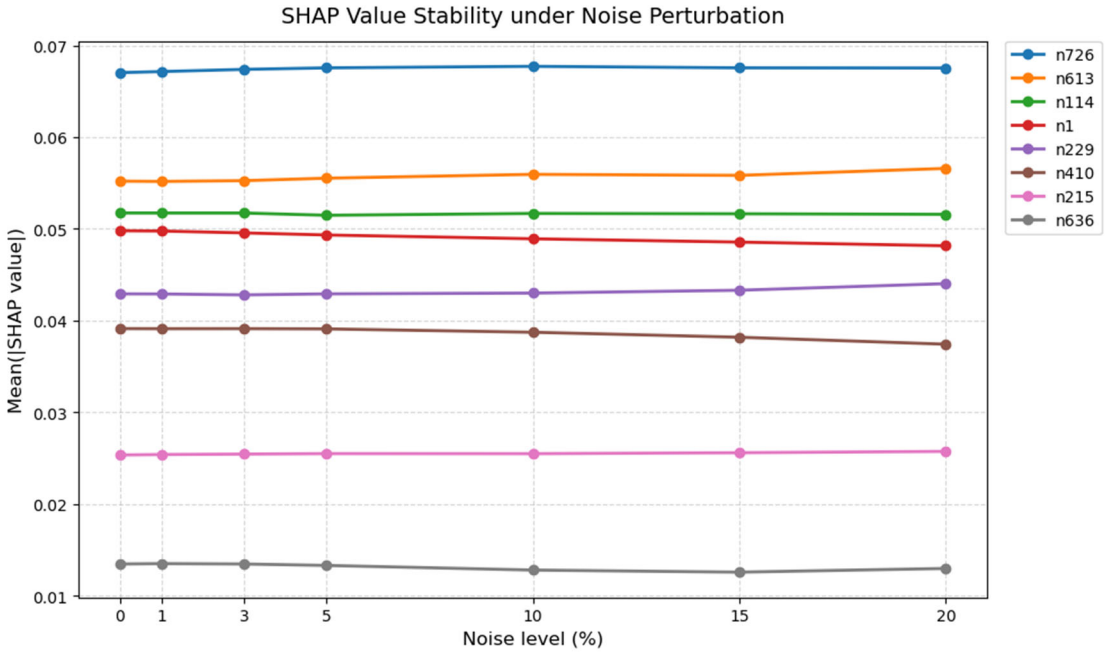


Figure 14. SHAP sensitivity analysis.

Figure 14 depicts the influence of noise perturbations on feature contributions, measured by mean absolute SHAP values across selected monitoring points. As the noise level increases from 0% to 20%, the SHAP values associated with each node display nearly horizontal trajectories with small variations. Moreover, the relative hierarchy of feature importance appears to remain largely consistent across all tested noise levels, with node n726 consistently providing the largest contribution and node n636 contributing the least. This indicates that the feature reliance and interpretability of the Stacking model are relatively stable under noise interference.

3.5. Extended Case Study

To further evaluate the generalizability of the proposed method and enhance the reliability of the conclusions, the Net3 example water distribution network provided in EPANET was selected for additional validation [45,46]. The network comprises 92 nodes, 117 pipes, 2 reservoirs, 3 water tanks, and 2 pumps. The monitoring points were determined based on the optimal sensor placement strategy proposed in [49]. Figures 15 and 16 present the topology of the Net3 network and the corresponding partitioning results after the addition of virtual nodes.

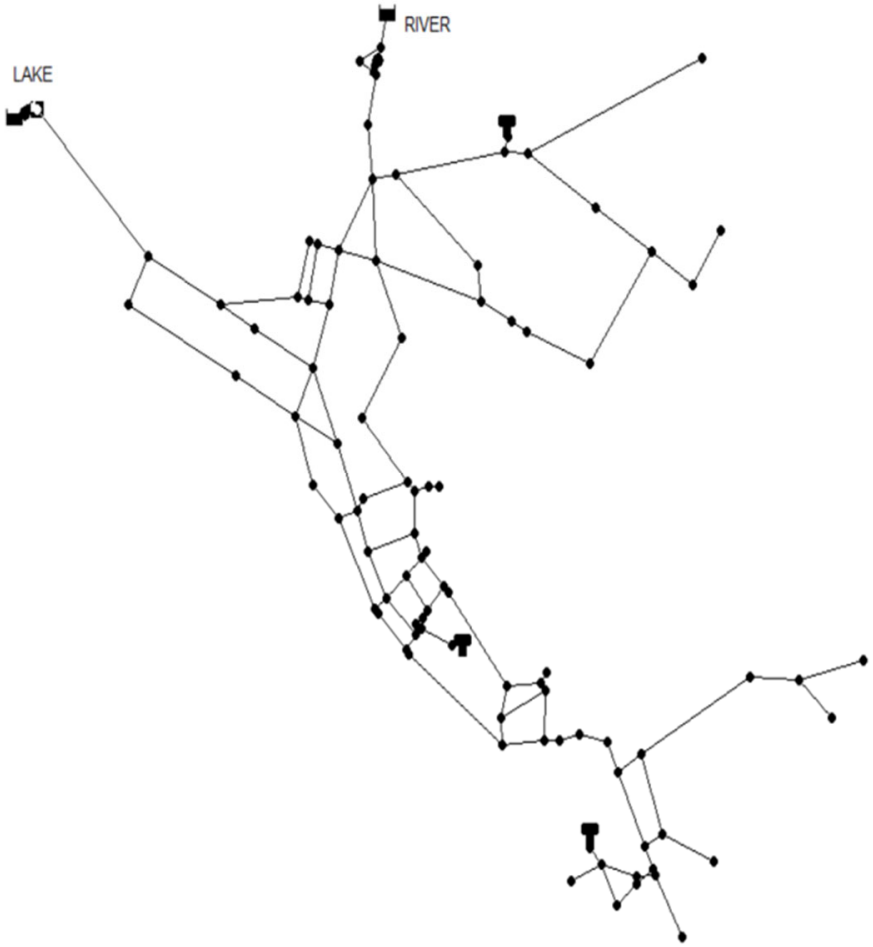


Figure 15. Topology of the Net3 water distribution network.

To further assess the generalizability of the proposed method, 2400 simulation samples were generated using the Net3 water distribution network. The same methodology was then applied to these samples. The performance of the evaluated models is summarized in Table 4, while the confusion matrix of the Stacking model is presented in Figure 17.

Table 4. Performance comparison on Net3.

Model	Accuracy	Macro-Precision	Macro-Recall	Macro-F1
RF	0.9729	0.9740	0.9729	0.9729
XGBoost	0.9708	0.9714	0.9708	0.9707
MLP	0.9708	0.9711	0.9708	0.9708
LightGBM	0.9729	0.9729	0.9729	0.9729
Voting	0.9771	0.9779	0.9771	0.9770
Stacking	0.9812	0.9819	0.9812	0.9812

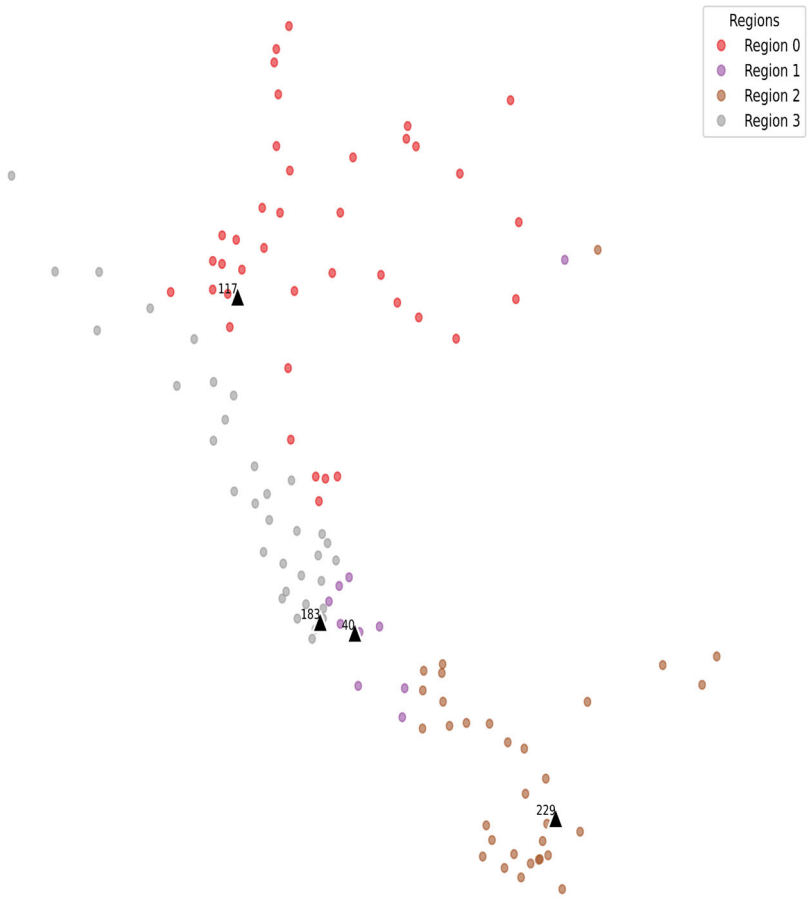


Figure 16. Partitioning results of the Net3 water distribution network.

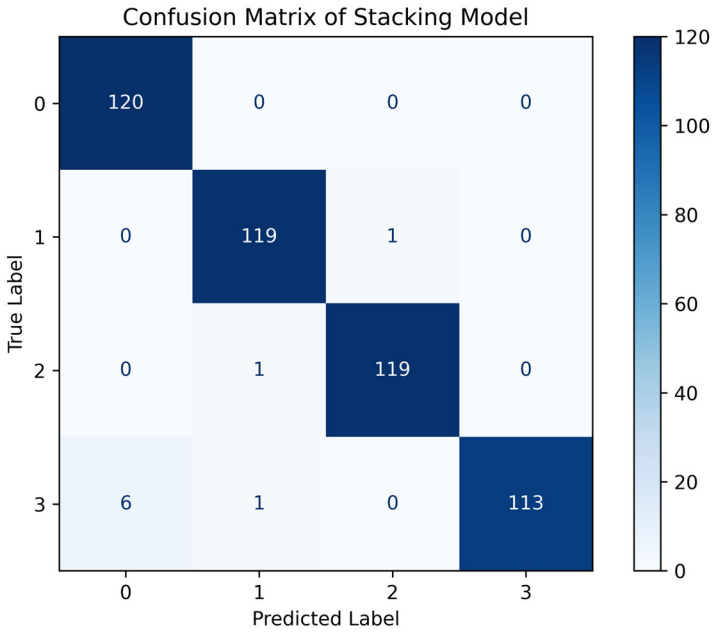


Figure 17. Confusion matrix of the Stacking model on Net3.

The results on the Net3 network demonstrate that the proposed method maintains good prediction performance on a different network topology. In particular, the Stacking model still achieves satisfactory performance, further demonstrating the generalizability and robustness of the proposed method.

4. Discussion

L-Town is a well-established and widely used benchmark network for leakage detection and localization in water distribution systems. Originally developed for the BattLeDIM competition, it has since been extensively adopted as a standardized testbed for evaluating and comparing different leakage detection and localization methods. For example, during the competition, the Cheng00 team employed change-point detection and statistical analysis for leakage detection and localization [50], the D and W team used pressure sensitivity analysis to identify potentially leaking pipes [51], and the Leakbusters team combined data-driven pressure reconstruction with hydraulic simulation [52]. Following the competition, a broader range of approaches have been investigated. Rajabi, et al. [53] proposed a method based on a Conditional Deep Convolutional Generative Adversarial Network (CDCGAN) and the Structural Similarity Index Measure (SSIM) for real-time leak detection and localization on L-Town, achieving approximately 70% detection accuracy, while Wang, et al. [54] developed a hydraulic-model-free approach based on graph theory and Bayesian reasoning, with localization errors below 300 m for all 10 leak points and below 70 m in some cases. Meanwhile, network partitioning has emerged as an important strategy for simplifying leakage localization. Dong, et al. [55] combined three-order embedding, K-means clustering, and Long Short-Term Memory (LSTM) for network partitioning and leak localization, incorporating pressure sensitivity, spatial information, and network connectivity, while Pérez-Sandoval, et al. [56] integrated Spectral Clustering with Bidirectional Long Short-Term Memory (Bi-LSTM) and achieved 85% effectiveness across 117 unseen leakage scenarios on L-Town.

Within this research landscape, the present study focuses on partitioning-based leakage localization while further addressing the prediction stage through an interpretable Stacking ensemble framework. Rather than depending on a single learning architecture, the proposed approach integrates RF, MLP, and XGBoost to exploit complementary predictive characteristics, and further reduces the sensing requirement through an optimized subset of eight pressure sensors. On the L-Town benchmark, the framework achieves an overall localization accuracy of 0.9667 and a Macro-F1 score of 0.9666, demonstrating strong localization performance. In a further test on the Net3 network, the same framework achieves an overall localization accuracy of 0.9812, providing additional evidence of the applicability of the proposed framework across different network configurations.

The integration of SHAP feature importance with confusion matrix analysis effectively bridges the gap between data-driven black-box predictions and physical hydraulic realities. The analytical results confirm that the model's decision-making logic aligns directly with network topology and sensor positioning rather than relying on arbitrary statistical artifacts. Distinct and spatially isolated monitoring points, such as node n1 at the northwestern extremity and node n114 in the southwest, exhibit highly concentrated SHAP contributions for their respective areas. This spatial isolation prevents cross-regional pressure signal overlap, allowing the ensemble model to achieve perfect localization accuracy in these boundary zones.

Conversely, instances of localized misclassification align closely with physical hydraulic boundaries. The symmetric misclassifications observed between Region 2 and Region 3 occur predominantly around shared high-conveyance hydraulic corridors. Monitoring points situated near these regional interfaces, such as node n726, register nearly identical pressure drop profiles during boundary leak events. This indicates that model confusion is not driven by algorithmic instability, but rather reflects genuine physical hydraulic overlap under sparse sensing constraints.

Regarding operational deployment, sensitivity analysis confirms that the Stacking model maintains reliable localization performance even under severe measurement noise,

retaining an accuracy above 0.85 under noise levels up to 20%. Crucially, the global SHAP feature importance rankings remain highly consistent across varying noise intensities, with critical hydraulic junction nodes like n726 and n613 consistently identified as the primary drivers of model predictions. This stability demonstrates that the combined spatial-hydraulic partitioning and non-redundant feature selection strategy effectively preserves the core signal-to-noise ratio necessary for practical real-time integration.

Nevertheless, several limitations should be acknowledged. First, the present evaluation was conducted primarily on the L-Town benchmark network, with the Net3 network used for additional validation. Although these two networks provide different network configurations and support the assessment of the generalizability of the proposed framework, they cannot fully represent the structural, hydraulic, operational, and sensor-related characteristics of real-world water distribution networks. Second, the leakage scenarios were generated based on predefined parameter ranges and simulation settings; therefore, uncertainties associated with actual leakage characteristics, demand variations, measurement errors, hydraulic parameter uncertainty, and operational changes may not be fully captured. Third, the proposed framework relies on pressure measurements from a limited number of monitoring points and on the partitioning strategy used to define the localization regions. Consequently, its performance may depend on sensor placement and network-specific hydraulic characteristics. Finally, although the results demonstrate strong performance on the evaluated benchmark networks, further validation using additional networks, field measurements, and experimentally or operationally observed leakage events is necessary to determine the robustness of the framework under diverse real-world conditions. Future research will therefore focus on extending the evaluation to a broader range of WDN topologies and operational conditions, incorporating greater uncertainty in hydraulic and leakage parameters, and validating the proposed framework using real-world monitoring data.

5. Conclusions

This study developed a spatial–hydraulic coupled and interpretable ensemble learning framework for leakage localization in urban WDNs. By combining spatial topology and hydraulic similarity, the framework delineated meaningful zones for leakage isolation, while the Stacking ensemble showed improved diagnostic performance over individual models and conventional Voting methods. SHAP analysis identified several monitoring nodes with substantial contributions to the regional classification, while the observed misclassifications were mainly concentrated between a small number of neighboring regions with similar feature representations. These results provide an interpretable view of the contributions of individual monitoring nodes to the classification process. An extended case study on the Net3 network further showed that the same framework achieved satisfactory localization performance under a different network configuration, providing additional evidence of its applicability across different WDN structures. Sensitivity analysis further indicated relatively stable diagnostic performance and feature-importance rankings under sensor noise of up to 20%. Overall, the framework shows potential for interpretable and robust leakage localization in practical WDN applications.

Despite these promising results, several methodological limitations remain to be addressed. First, the framework was validated using the synthetic L-Town benchmark network under nighttime steady-state conditions, assuming single-partition leakage scenarios without considering real-time demand dynamics or multiple simultaneous leaks. Second, model development relied on simulated EPANET data, which may not fully reflect sensor drift, missing data, or dynamic topology changes encountered in real operational environments.

To overcome these limitations, future research will focus on expanding the framework to accommodate unsteady-state daytime hydraulics, non-stationary demand fluctuations,

and complex scenarios involving concurrent leakage events across multiple partitions. Additionally, incorporating domain-specific physical constraints directly into the ensemble architecture and testing against real-world operational datasets will be crucial. Future efforts will also explore adaptive sensor placement algorithms and transfer learning methodologies to facilitate seamless deployment across heterogeneous WDN topologies without retraining from scratch. From an operational perspective, the proposed framework offers significant practical relevance for water utility management. By translating black-box machine learning outputs into transparent, SHAP-guided spatial diagnostics, utility operators can rapidly isolate leakage-prone zones, optimize pipe repair schedules, and minimize non-revenue water (NRW) losses. Furthermore, the non-redundant sensor feature selection methodology provides a cost-effective strategy for utilities seeking to upgrade sparse monitoring infrastructures under strict capital expenditure constraints.

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